

Predicting Ligue 1 Match Outcomes with Player-Level Statistics

Data Collection & Feature Organisation

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Ligue 1 · Seasons 2018–2024 · 1,653 matches · 73 features

Outline

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- 4 Feature Engineering
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Research Question

Do **individual football player** rolling statistics add predictive signals beyond team-level aggregates?

Goal: predict Ligue 1 match outcomes (home win, draw, away win) using team and player data. Train on 2018–2022; test on 2023/24.

Three feature regimes evaluated:

Regime A

Team-level aggregates
29 features

Regime B

Team + FBref match
statistics
50 features

Regime C

Full set incl. **player-rolling**
features
76 features



DB FBref

What: Team-level match records and player-season summaries

Seasons: 2018–2023

Rows: 62,683 player-match rows

Script: `collect_fbref.py`

Understat

What: Match results, xG values, and pre-match forecast probabilities

Seasons: 2018–2023

Rows: 2,105 matches

Script:
`collect_understat.py`

DB Transfermarkt

What: Estimated squad market values (euros) and average player age

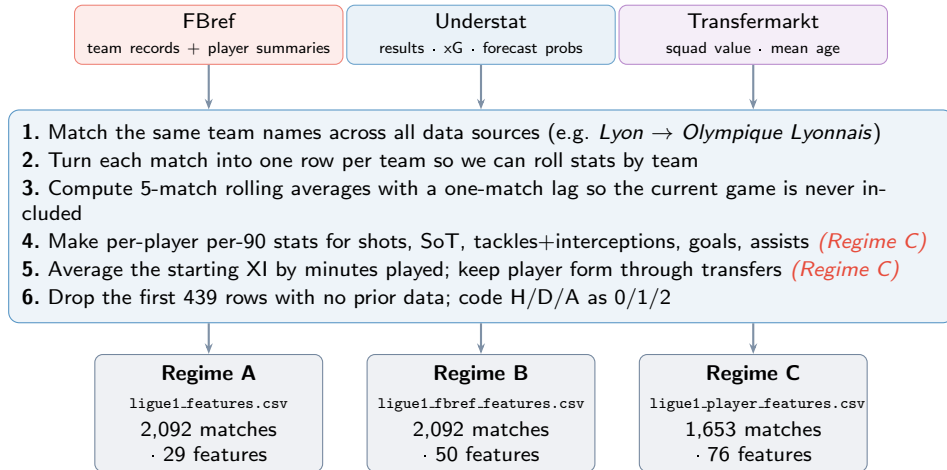
Seasons: 2018–2023

Rows: 174 team-seasons

Script:
`collect_transfermarkt.py`

FBref data was extracted via a scripted pipeline; Understat data via its API; Transfermarkt values from a reliable GitHub Repo. The 2021/22 FBref season required manually completing a CAPTCHA before resuming the scripted download.

Data Pipeline





Feature regimes

Regime	Matches	Features
A: team only	2,092	29
B: +FBref	2,092	50
C: +player	1,653	76

Regime C loses 439 rows from cold-start roll-up.

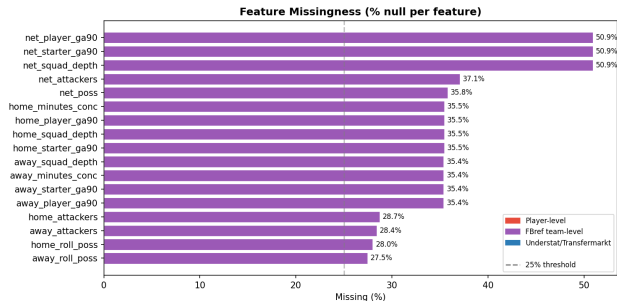
Features (Regime C)

Source	Features
Understat / Transfermarkt	29
FBref team-level	21
Player-level	26
Total	76

Season breakdown

Season	Matches	Note
2018/19	370	COVID curtailed
2019/20	278	
2020/21	379	
2021/22	379	CAPTCHA step
2022/23	380	Partial (May 2024)
2023/24	306	
Total	2,092	

Missing Data: Sources & Resolution



What is missing?

- No FBref *team* data for 2021/22 – not able to collect team-level data, but player-level was captured
- Rolling features need prior matches — first ≈ 5 matchdays per season are null (cold-start)
- net.* at $\sim 51\%$: null if *either* side null

What we did

- No imputation – null rows dropped at fit time
- Same subset used across all regimes
- Regime C: 439 cold-start rows dropped $\rightarrow$ 1,653 matches

What this means

- 2021/22 excluded equally from A, B, C – no regime penalised
- Regime C subset smaller – note when comparing across regimes
- Understat / Transfermarkt: zero missingness



Source	Features engineered	Regimes
Understat + Transfermarkt	Rolling xG, goals, wins, points; squad value, age	A, B, C
FBref team-level	Rolling possession, formation, goals + assists per 90 (team)	B, C
Player-level	Rolling shots, shots on target, defensive actions, goals, assists per 90	C only

- **Anti-leakage:** 5-match rolling window with `shift(1)` — current match never in its own features
- **Bookmaker probabilities excluded** from model inputs — kept as external baseline only (including them would be circular)
- **Player rolling stats are personal, not club-based** — if a player transfers, their form history carries over



Each player's 5-match rolling stat is aggregated across the starting XI, **weighted by minutes played**:

$$\hat{s}_{\text{team}} = \frac{\sum_{p \in \text{XI}} \text{min}_p \cdot \text{roll}_p}{\sum_{p \in \text{XI}} \text{min}_p}$$

Each stat then becomes three features:

home_pl_* away_pl_* net_pl_*

Stats tracked per player

Feature	Meaning
sh_p90	shots / 90
sot_p90	shots on target / 90
def_p90	tackles + int. / 90
gls_p90	goals / 90
ast_p90	assists / 90
starter_mins	minutes regularity
pct_fw	fraction forwards
depth_sh	starter vs bench gap



1. Filter Method

Simple idea: if I know this feature, do I get a better clue about the match result? Good for catching patterns that are not just straight lines.

2. Wrapper Method

Start with many features, remove the weakest ones step by step, and keep the set that gives the best validation score. Best subset: **26 features**.

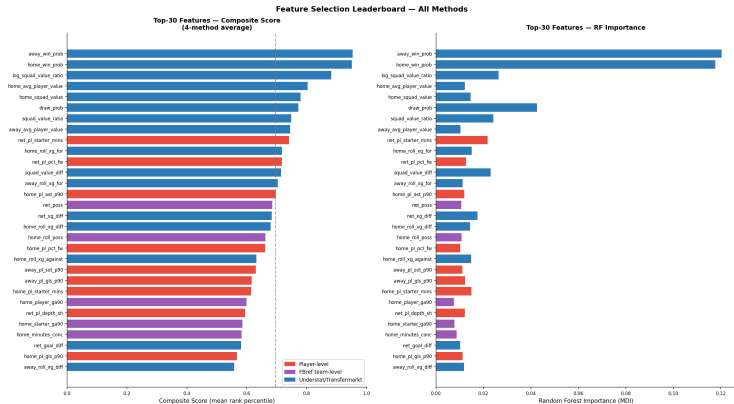
3. Embedded Method — Lasso

A linear model with a penalty that pushes unhelpful features down to exactly zero. 17 features were removed; **0 player features** were.

4. Embedded Method — Decision Tree

Let a forest of trees rank which features it uses most to make good splits. This gives a second opinion that broadly matches RFECV.

Feature Leaderboard: Top 30 Features

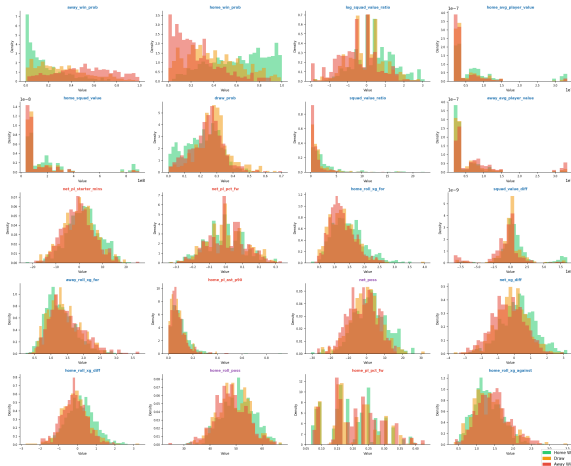


■ Understat/Transfermarkt ■ FBref team-level ■ Player-level

Feature Distributions by Match Outcome (Top 20)



Feature Distributions by Match Outcome — Top-20 Features
(title colour: red=player-level, purple=fbref-team, blue=Understat)



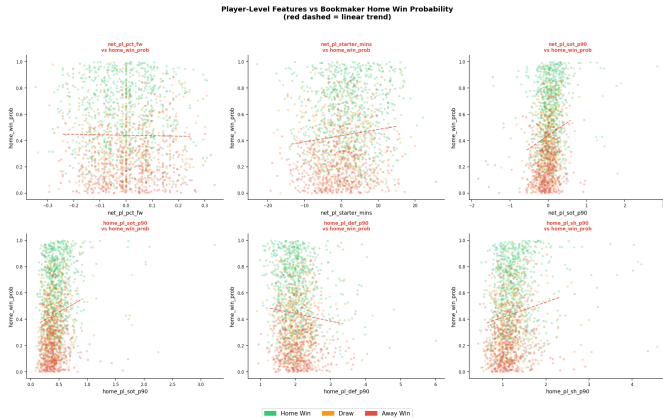
Each histogram shows the distribution of one feature split by match outcome (H/D/A).

Less overlap between the three coloured distributions $\Rightarrow$ more discriminative feature.

Bookmaker probability features show the clearest separation.

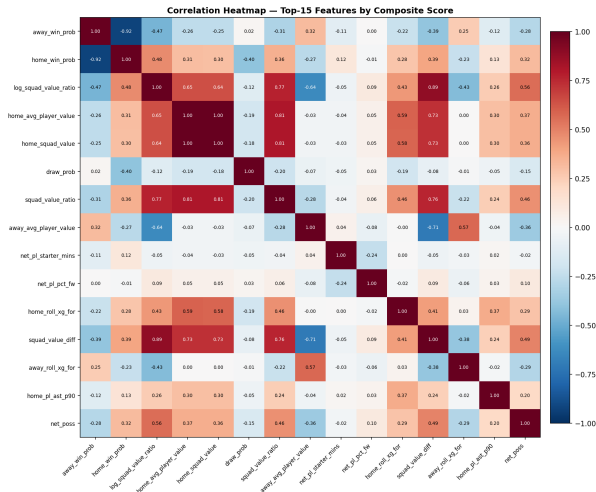
Player-level features add a small but consistent nudge on top of the bookmaker signal.

Bookmaker Odds vs Player-Level Features



The bookmaker's implied probability and the player rolling stats **mostly disagree** — wide scatter, no tight trend.

Correlation Heatmap: Top 15 Features



Red zones = high correlation $\Rightarrow$ redundant (e.g. home_win_prob and away_win_prob).

Player features show near-white rows and columns $\Rightarrow$ barely correlated with anything else.

This confirms they bring **genuinely new information** rather than repeating the bookmaker signal.



Model setup

Algorithm: multinomial logistic regression
(softmax output)

Library:

`sklearn.linear_model.LogisticRegression`

Hyperparameters: $C=1.0$ (default L2),
`solver=lbfgs`, `max_iter=1000`

Scaling: `StandardScaler` fit on *train only*

Protocol

Split: train 2018–2022 / test 2023/24

Common subset: 593 matches with complete data across all 3 CSVs

Understat excluded from features (used as external baseline only)

3 regimes: team (29) / +FBref (50) / +player (76)

Why exclude Understat from the features?

Including the baseline as a model input would be **circular** — we'd partly be measuring what Understat already knows, not what our engineered features add.

Linear Baseline — Results



Test results on the held-out 2023/24 season (168 matches).



Log-loss by regime

Regime	Log-loss
A: team (29)	1.06
B: +FBref (50)	1.14
C: +player (76)	1.22
Uniform reference	1.099
Understat baseline	0.910

Brier (A): 0.62 · Acc. (A): 49.2%

Log-loss *increases* with more features — the central diagnostic of the project.

Linear Baseline — Limits & Diagnostic



Why did adding features hurt? Two compounding problems explain it.

1. Multicollinearity

Team-level and player-level features measure overlapping things (a team's xG is essentially a function of its players' xG). Linear models cannot decide where to attribute the weight — the optimisation has multiple equivalent solutions, leading to **unstable coefficients**.

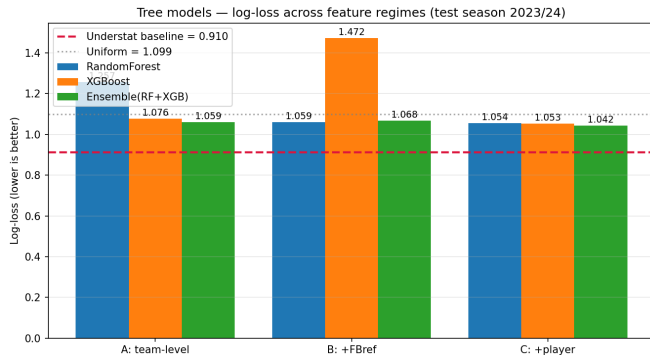
2. Curse of dimensionality

Only **417 training matches** against up to **76 features** — a samples-to-features ratio of ~ 5.5 . Rule of thumb: linear models need 10–20 $\times$ more samples than features. Below that, they **overfit the noise** rather than learn the signal.

This negative result is what motivated the tree models

Tree-based models handle correlated features natively — each split uses only one feature at a time, so they're *locally low-dimensional*. **Tree models flip the trend** (RF: 1.26 $\rightarrow$ 1.06 $\rightarrow$ 1.05). Same data, same protocol, opposite outcome — which Samuel will detail next.

Lesson: *model class matters more than feature count. The player-level signal exists — but linear models cannot surface it.*



Test log-loss (season 2023/24)

	A	B	C
RF	1.26	1.06	1.05
XGB	1.08	1.47	1.05
Ensemble	1.06	1.07	1.04
LR (ref)	1.06	1.14	1.22

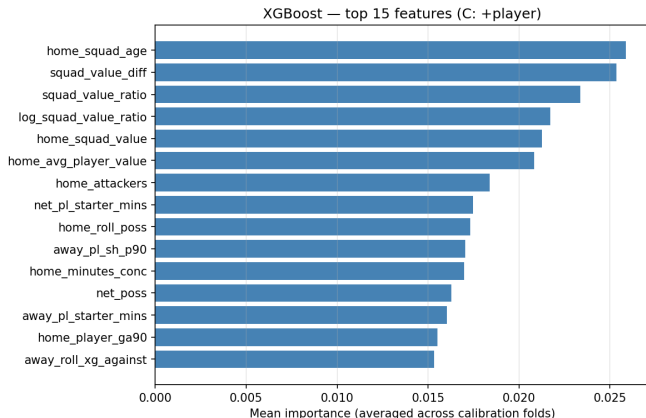
A=team, **B**=+FBref, **C**=+player

Accuracy in context (3-class):

Random guess	33%
Always home	43%
Best model	50%
Bookmaker	56%

No model clears the bookmaker baseline (0.91 log-loss). Bookmakers have real-time info (injuries, team news) that rolling historical stats cannot capture.

Modeling — Feature Importance (XGBoost, full regime)



How to read: mean importance over the calibration folds for the best XGBoost on the full player regime.

Team-level form (rolling xG, market value) sits at the top, as expected.

Player-level features (`starter_mins`, `pct_fw`, `depth_gaps`) appear repeatedly in the top ranks — they are not just noise the model tolerates, they are signal the model actively uses.

Consistent with the feature-selection leaderboard (`net_pl_starter_mins` rank 9, `net_pl_pct_fw` rank 11).



1. Tree models flip the feature-regime story

Logistic regression *got worse* as features were added ($1.06 \rightarrow 1.14 \rightarrow 1.22$): multicollinearity and curse of dimensionality. Tree models *improve* with more features (RF: $1.26 \rightarrow 1.06 \rightarrow 1.05$) — they handle correlated inputs natively.

2. Player features add real signal

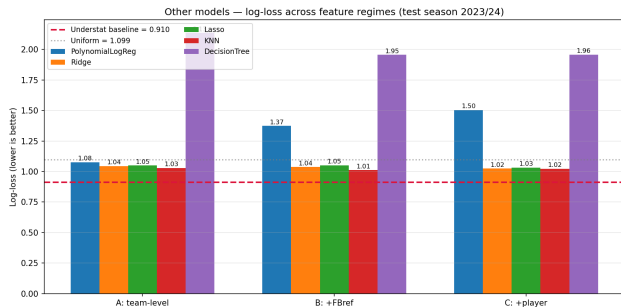
The full player regime (C) is the best for every tree model. Confirms the project hypothesis: player-level rolling stats add predictive power once the model can use them.

3. Ensembling extracts a final gain

RF and XGB make different mistakes; averaging their calibrated probabilities beats either alone (1.04 vs $1.05/1.05$ in regime C).

Other Models — Linear, Polynomial, KNN, Decision Tree

Same protocol, same temporal holdout, same 3 regimes. We tested four additional model families to see whether the choice of algorithm actually matters at our sample size.



Test log-loss (season 2023/24)

	A	B	C
Polynomial	1.08	1.37	1.50
Ridge (L2)	1.04	1.04	1.02
Lasso (L1)	1.05	1.05	1.03
KNN	1.03	1.01	1.02
Decision Tree	2.13	1.96	1.96
RF+XGB ens.	1.06	1.07	1.04

A=team, **B**=+FBref, **C**=+player

Six algorithms cluster within 0.03 log-loss — we've hit the *information ceiling* of post-match aggregate stats. Closing the gap to Understat (0.91) needs richer features, not a different model.

Per-model takeaways: **Polynomial** overfits as features grow ($d=2$ explodes to 400^+ inputs); **Ridge/Lasso** match each other — regularisation type doesn't matter much here; **KNN** surprises by tying the tree ensemble (small dataset $\rightarrow$ simple wins); **Decision Tree** confirms the value of bagging/boosting — a single unregularised tree is the worst model in every regime

KNN — Choosing k

k was not hand-picked. The full grid $k \in \{5, 10, 15, 20, 30, 50\} \times \{\text{uniform, distance}\} \times \{\text{Manhattan, Euclidean}\}$ was searched under the same `TimeSeriesSplit(5)` protocol and the lowest-log-loss combination was kept.

Why those bounds?

MIN $k = 5$: smaller k gives degenerate probabilities (denominators of 2–3 only output $\{0, 0.33, 0.5, 0.66, 1\}$) — log-loss punishes that hard.

MAX $k = 50$: the smallest CV training fold is ~ 75 rows; sklearn requires $k \leq$ fold size. We initially tried $k = 100$ and got runtime warnings, so we capped at 50.

Winners (per regime)

Regime	k	metric
A: team	50	Euclidean ($p=2$)
B: +FBref	50	Manhattan ($p=1$)
C: +player	50	Manhattan ($p=1$)

All three regimes **independently** picked $k = 50$ with distance-weighting. The Minkowski exponent drifted from $p=2$ to $p=1$ as features grew — consistent with the textbook result that L1 is more robust in higher dimensions.

Why $k = 50$ makes domain sense: distance-weighting means the closest 5–10 historical matchups still drive the prediction; the further 40 just provide a regularising prior. Football is noisy, so averaging $\sim 12\%$ of the historical sample smooths randomness without washing out the matchup-specific signal.



- ① Player-level rolling statistics add real predictive signal beyond team form and bookmaker odds.
- ② Feature selection backs that up: `net_pl_starter_mins` and `net_pl_pct_fw` rank highly across all four methods, and RFECV keeps 10 player features in its final subset.
- ③ Tree models give the clearest proof of the player signal: as we move from team-only features to the full player regime, every tree model improves, while logistic regression gets worse under the extra correlation and dimensionality.
- ④ The best tree result is the RF + XGBoost ensemble on the full player regime at **1.04** log-loss and 50.6% accuracy, so ensembling still extracts a small final gain.
- ⑤ KNN is a useful surprise: with $k=50$ it ties the stronger models (**1.01** in regime B, 1.02 in regime C), suggesting that averaging similar past match states works well when the sample is modest and noisy.
- ⑥ Six different algorithms finish within about 0.03 log loss of one another, which points to